Lab 4: Common Emitter Amplifier

A common-emitter amplifier is a single-stage transistor circuit that amplifies an input signal. It is called "common-emitter" because the emitter terminal is common to both the input and output circuits, which means the input is applied to the base, and the output is taken from the collector. This configuration offers a high voltage gain and high current gain, and it inverts the output signal (180° out of phase with the input).

Lab Objective

The purpose of this experiment is to demonstrate the operational characteristics of a small signal common emitter amplifier with a 180 degree phase shift. The input is applied to the base and the output is measured from the collector.

Apparatus List

Apparatus		Quantity
1.	NPN Transistors	1
2.	Resistors:	
	560 Ω, 820 Ω, 2.2 kΩ, 10 kΩ, 47 kΩ	1 each
3.	Capacitors:	
	10 uF, 22 uF and 3.3 uF	1 each
4.	Power supply	1
5.	Oscilloscope	1
6.	Function generator	1

Lab procedure

1. Connect the common emitter circuit as shown in Fig. 4.1.

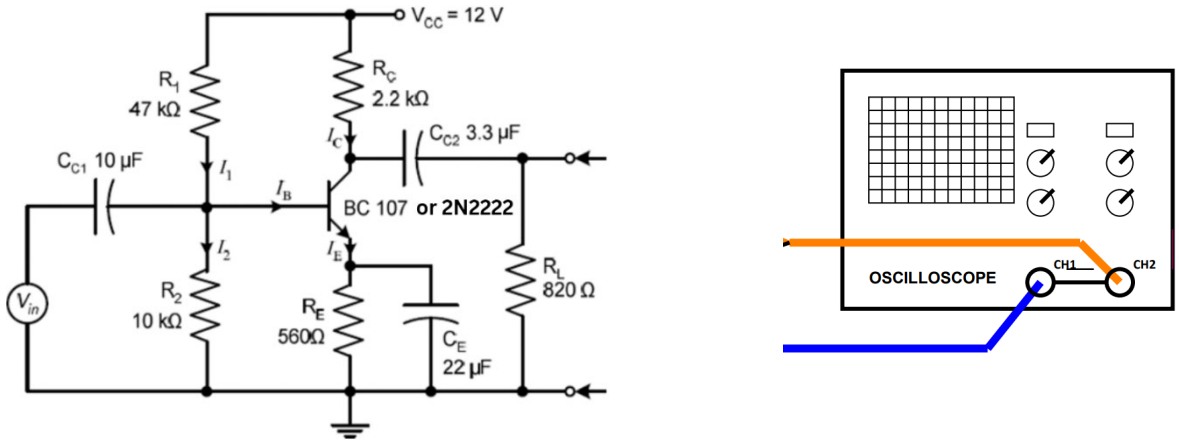


Fig. 4.1 A fully bypassed common emitter circuit.

Lab procedure (cont.)

2. Generate sinusoidal waveform from function generator with amplitude = 10 mV at 1 kHz to be used as input signal V_{in}.
3. Use channel 1 of the oscilloscope to capture the waveform of V_{in} and use channel2 to capture the waveform of output voltage across load resistance (820 Ω).
4. Plot both waveforms on the same graph sheet.
5. Find voltage gain ($A_v = \frac{v_{out}}{v_{in}}$).
6. If load resistance (820 Ω) is removed, how it affects the voltage gain?

Lab result

Voltage gain	No load	With load
$A_v = \frac{v_{out}}{v_{in}}$		

Reference

2N2222 / 2N2222A

NPN Silicon Epitaxial Planar Transistor
for switching and AF amplifier applications.



1. Emitter 2. Base 3. Collector
TO-92 Plastic Package

Characteristics at T_a = 25 °C

Parameter		Symbol	Min.	Max.	Unit
DC Current Gain					
at V_{CE} = 10 V, I_C = 0.1 mA		h_{FE}	35	-	-
at V_{CE} = 10 V, I_C = 1 mA		h_{FE}	50	-	-
at V_{CE} = 10 V, I_C = 10 mA		h_{FE}	75	-	-
at V_{CE} = 10 V, I_C = 150 mA		h_{FE}	100	300	-
at V_{CE} = 10 V, I_C = 500 mA	2N2222	h_{FE}	30	-	-
	2N2222A	h_{FE}	40	-	-
Collector Base Cutoff Current					
at V_{CB} = 50 V	2N2222	I_{CBO}	-	10	nA
at V_{CB} = 60 V	2N2222A		-	10	
Collector Base Breakdown Voltage					
at I_C = 10 µA	2N2222	$V_{(BR)CBO}$	60	-	V
	2N2222A		75	-	
Collector Emitter Breakdown Voltage					
at I_C = 10 mA	2N2222	$V_{(BR)CEO}$	30	-	V
	2N2222A		40	-	
Emitter Base Breakdown Voltage					
at I_E = 10 µA	2N2222	$V_{(BR)EBO}$	5	-	V
	2N2222A		6	-	
Collector Emitter Saturation Voltage					
at I_C = 150 mA, I_B = 15 mA	2N2222	$V_{CE(sat)}$	-	0.4	V
	2N2222A		-	0.3	
at I_C = 500 mA, I_B = 50 mA	2N2222		-	1.6	
	2N2222A		-	1	
Base Emitter Saturation Voltage					
at I_C = 150 mA, I_B = 15 mA	2N2222	$V_{BE(sat)}$	-	1.3	V
	2N2222A		0.6	1.2	
at I_C = 500 mA, I_B = 50 mA	2N2222		-	2.6	
	2N2222A		-	2	